

CPSC-406 Report

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February 15, 2026

Abstract

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1 Introduction

2 Week by Week

2.1 Week 1

Exercise 1

Consider the following DFAs $\mathcal{A}_1$ and $\mathcal{A}_2$.

Complete the following table with all 8 possible strings of length 3 over $\{a, b\}$:

String	Accepted by $\mathcal{A}_1$?	Accepted by $\mathcal{A}_2$?
aaa	No	Yes
aab	Yes	Yes
aba	Yes	No
abb	Yes	No
baa	No	Yes
bab	No	No
bba	No	No
bbb	No	No

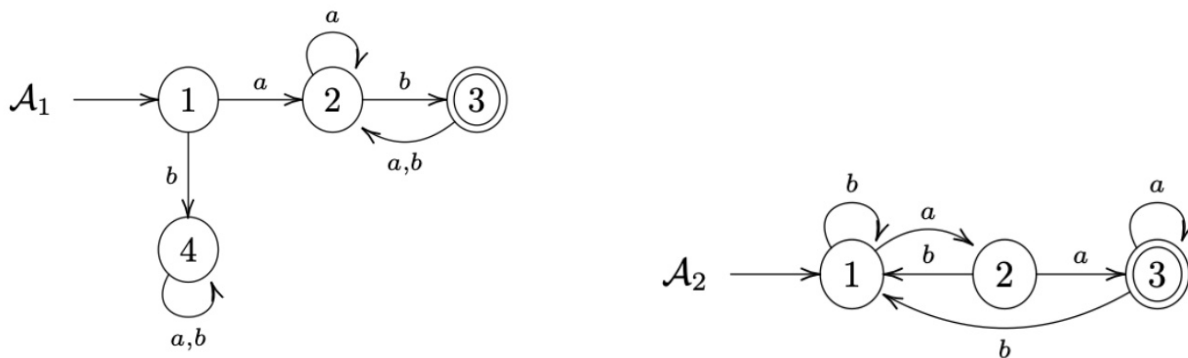


Figure 1: DFAs $\mathcal{A}_1$ (left) and $\mathcal{A}_2$ (right).

Exercise 1.2

describe the languages $L(\mathcal{A}_k)$ accepted by $\mathcal{A}_k$ for $k = 1, 2$.

Answer:

$L(\mathcal{A}_1)$: The language accepted by $\mathcal{A}_1$ consists of all strings over $\{a, b\}$ that start with a and contain at least one b . Starting with b leads to the trap state 4 and is rejected. $L(\mathcal{A}_2)$: The language accepted by $\mathcal{A}_2$ consists of all strings over $\{a, b\}$ that contain the substring aa . Reading two consecutive a 's leads to the accepting state 3.

Exercise 2 (Designing DFAs)

Design DFAs whose accepted languages are given as follows:

1. All the words that end with ab
2. All the words that contain aba
3. All the words that contain an odd number of a 's and an odd number of b 's
4. All the words that contain an even number of a 's and an odd number of b 's
5. All the words such that any three consecutive characters contain at least one a
6. All the words that contain bbb

What do you notice when comparing the various automata?

Answer: Several patterns emerge when comparing the six DFAs. First, they all share the same alphabet $\{a, b\}$ and differ mainly in the kind of property they check. Some automata focus on *local* conditions (substrings like ab , aba , or bbb), while others track *global* properties (parity of a 's and b 's). The substring-based DFAs (1, 2, 6) use states to remember how much of the target pattern has been seen and typically need a trap state for “pattern not found.” The parity-based DFAs (3, 4) use a fixed structure with four states representing the four combinations of (even/odd a , even/odd b), showing that such languages have a uniform, predictable form. The sliding-window constraint (5), where every block of three consecutive symbols must contain an a , is more involved because it requires the automaton to encode what was seen in the last few positions. Overall, the same formal model (DFA) can express very different language types, from simple suffix checks to parity counting to bounded-memory pattern constraints.

Thought process. I decided to map out a truth table and filled in all of the conditions and found that the outputs were satisfied at every condition, leading me to the conclusion that every transition was a satisfactory state condition.

3 Synthesis

4 Evidence of Participation

5 Conclusion

References

[BLA] Author, [Title](#), Publisher, Year.